Question 2: Fine-Tuning RoBERTa for 5-Class Amazon Review Rating Prediction

DSL 504 Natural Language Processing, Group Assignment 2

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Abstract

We predict the 1–5 star rating of English Amazon reviews with two models: a BiGRU with additive attention pooling over GloVe embeddings, and a fully fine-tuned `roberta-base` with masked mean pooling. Over two seeds, RoBERTa reaches **66.3% accuracy, 66.1% macro-F1 and 0.378 MAE** on the test set against 63.3%, 62.8% and 0.437 for the BiGRU. RoBERTa improves every class, most of all the ambiguous 2★ and 3★, and halves the number of 1↔5-star confusions; its gain over the baseline comes mainly from fewer errors of two or more stars, while confusions between adjacent ratings such as 4★ vs 5★ barely change. In the ablation, first-token pooling matches mean pooling (66.4% vs 66.1% test macro-F1, within about one seed standard deviation).

1 Task, data and protocol

The data is `SetFit/amazon_reviews_multi_en`: 200,000 training, 5,000 validation and 5,000 test reviews, perfectly balanced (40,000 training and 1,000 validation/test reviews per star). Labels 0–4 are stars -1, so MAE is in stars.

Titles. The SetFit copy keeps only the review body. `mteb/amazon_reviews_multi` (config `en`) holds the same reviews with the same ids and splits, stored as *title*`\n\n`*body*. Splits and labels are taken from SetFit as the brief asks; each title is matched by review id and attached only if the two bodies agree after whitespace normalisation, which held for 100% of reviews in all three splits. The model input is `"<title>. <body>"` (a full stop is added when the title has no end punctuation).

Protocol. Validation is used for early stopping, checkpoint selection and the ablation. The test split is loaded in exactly one place (`04_evaluation.ipynb`): every run predicts it once, the predictions are saved, and all test numbers, figures and examples in this report are computed from those saved predictions. Nothing was retrained or tuned after the test predictions existed.

2 Preprocessing: two pipelines

	Classic (BiGRU)	**RoBERTa**
Cleaning	strip HTML tags and URLs, Unicode NFC, lowercase (GloVe 6B is uncased)	none (the tokenizer is cased and byte-level)
Tokens	spaCy rule-based English tokenizer (`spacy.blank("en")`)	`RobertaTokenizerFast`, byte-level BPE, `<s>` … `</s>`
Vocabulary	from the training split only, min. frequency 2: 32,878 types incl. `<pad>` = 0, `<unk>` = 1; test OOV rate 0.39%	fixed 50,265-piece vocabulary; no unknown tokens
Embeddings	GloVe 6B 300-d: vectors for 88.1% of types and 99.7% of training tokens; others $\sim \mathcal{N}(0, \sigma_{\text{GloVe}})$, `<pad>` zero	pre-trained `roberta-base`
Length	median 34, 95th pct. 117 tokens; truncated to **200** keeping the head (and so the title); 0.93% of training reviews are longer	median 38, 95th pct. 124; truncated to **256** keeping `</s>`; 0.44% longer
Batching	padded per batch with `<pad>`, lengths passed on for packing	`attention_mask`; dynamic padding with `DataCollatorWithPadding` (to a multiple of 8)

Table 1: The two input pipelines (statistics on the training split unless stated).

Both pipelines tokenise once and then pad per batch. Training batches are drawn by a seeded length-grouped sampler (shuffled pools of similar-length reviews, batch order shuffled), which keeps padding low; evaluation keeps the split's order so predictions line up with review ids.

3 Models

BiGRU + additive attention (baseline, 12.0 M parameters, 9.9 M of them in the embedding). 300-d embedding initialised from GloVe → 2-layer bidirectional GRU, 256 units per direction, dropout 0.3 between layers, on packed sequences so neither direction reads padding → additive (Bahdanau-style) attention pooling → dropout 0.3 → linear 512 → 5. With GRU states h_t and mask m_t (1 for real tokens):

$$e_t = v^\top \tanh(W h_t + b), \qquad \alpha_t = \frac{m_t \exp e_t}{\sum_s m_s \exp e_s}, \qquad c = \sum_t \alpha_t h_t.$$

The embedding is frozen in epoch 1 and fine-tuned from epoch 2.

RoBERTa (124.1 M parameters, all trainable). `RobertaModel` (`roberta-base`) → masked mean of the last hidden layer over every non-padding token (`<s>` and `</s>` included; the sum is taken in fp32) → dropout 0.1 → linear 768 → 5: $p = \sum_t m_t H_t / \sum_t m_t$. The ablation variant uses $p = H_{\texttt{<s>}}$ instead.

4 Training

	BiGRU	**RoBERTa**
Loss, precision	cross-entropy; fp16 autocast with gradient scaling	cross-entropy; fp16 autocast with gradient scaling
Optimiser	Adam, lr 10^{-3}, gradient clipping at 1.0	AdamW, lr 2×10^{-5}, weight decay 0.01 (none on biases and LayerNorm), clipping at 1.0
Schedule	constant	linear warmup over 6% of steps (750 of 12,500), then linear decay to 0
Batch	128	32, split over two T4 GPUs (`DataParallel`); gradient accumulation supported but not needed
Selection	validation after every epoch; early stopping on validation macro-F1, patience 3, at most 15 epochs; best epoch restored	2 epochs, validation every half epoch; checkpoint with the best validation macro-F1 kept
Outcome	best epoch 3 for both seeds, stopped after epoch 6; 3.5 min per run on one T4	best checkpoint at epoch 2.0 for all four runs; 43 min per run on two T4s

Table 2: Training setup. Seeds 42 and 43 for every configuration; the seed is set before building the model and the data loaders, and the batch sampler and loader workers are seeded too.

A library version matters. With `transformers` 5.16.1 (Kaggle's preinstalled version) the same 20k-review RoBERTa check run reached only 0.49 validation macro-F1, against 0.64 with 5.20.2; switching the attention implementation, fp16 vs fp32, and one vs two GPUs made no difference. All reported runs use 5.21.0, and the code refuses to train on anything older than 5.20.2.

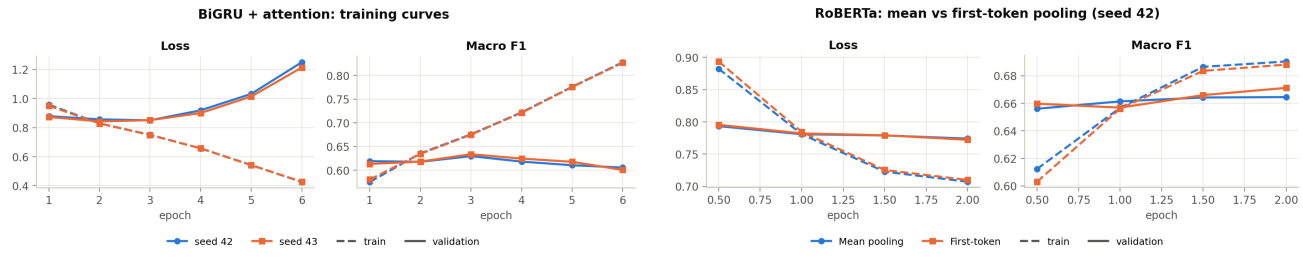


Figure 1: Training (dashed) and validation (solid) loss and macro-F1. Left: BiGRU, both seeds, per epoch. Right: RoBERTa mean vs first-token pooling (seed 42), every half epoch; train values average the batches since the previous evaluation.

Curves. The BiGRU's validation macro-F1 peaks at epoch 3 and its validation loss rises from epoch 4 while training F1 climbs to 0.83: once the 9.9 M embedding parameters are unfrozen the model memorises the training set quickly, so early stopping is essential. RoBERTa's validation loss is still falling slowly at epoch 2 with a much smaller train–validation gap, and every run's best checkpoint is its last one, so a third epoch might add a little (see Limitations).

Model	Validation			Test		
	Accuracy (%)	Macro-F1 (%)	MAE (stars)	Accuracy (%)	Macro-F1 (%)	MAE (stars)
BiGRU + attention	63.55 ± 0.27	63.19 ± 0.28	0.434 ± 0.003	63.30 ± 0.20	62.82 ± 0.16	0.437 ± 0.002
RoBERTa, mean pooling	66.64 ± 0.08	66.41 ± 0.06	0.377 ± 0.002	**66.31** ± 0.01	**66.11** ± 0.02	**0.378** ± 0.001
RoBERTa, first token	67.11 ± 0.33	66.91 ± 0.30	0.370 ± 0.004	66.54 ± 0.17	66.36 ± 0.18	0.376 ± 0.002

Table 3: Mean ± sample standard deviation over seeds 42 and 43, at each run's best validation checkpoint. RoBERTa with mean pooling is the model the brief specifies; first-token pooling is the ablation.

5 Results

RoBERTa beats the BiGRU by **+3.0 accuracy, +3.3 macro-F1 and −0.059 MAE** on the test set, gaps 10–20 times larger than the seed-to-seed spread. Test scores sit within 0.6 points of validation for every model, so selecting checkpoints on validation did not overfit it. Per class (Fig. 2, left), RoBERTa is better everywhere, most on 2★ (+4.6 F1) and 3★ (+4.4) and least on 5★ (+1.9) and 4★ (+2.3). The middle ratings are the hardest for both models (52–57 F1) because these reviews mix praise and criticism and the line between "disappointing" and "average" is fuzzy even for people.

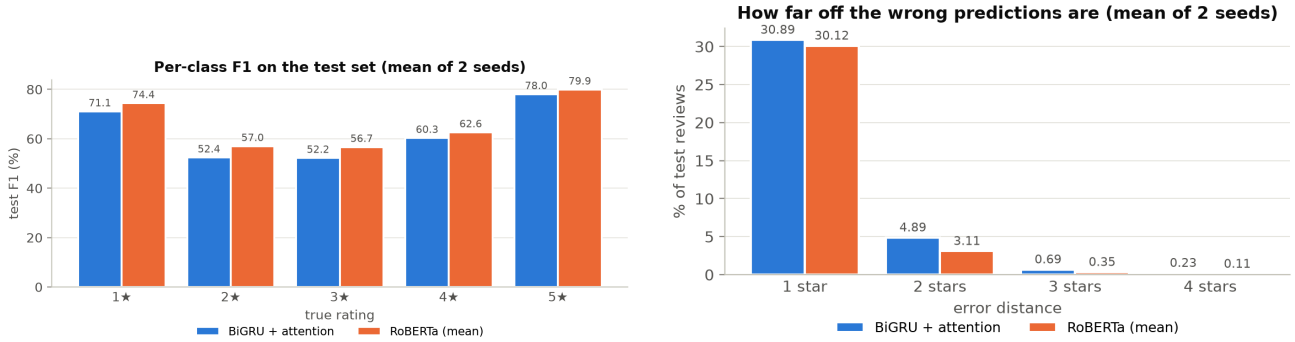


Figure 2: Test set, mean of two seeds. Left: per-class F1. Right: share of all test reviews predicted 1, 2, 3 or 4 stars away from the truth.

6 Ablation: mean vs first-token pooling

Only the pooling changes; data, seeds, schedule and every other setting are identical. On validation, first-token pooling is 0.50 points of macro-F1 ahead (66.91 ± 0.30 vs 66.41 ± 0.06); on test the lead shrinks to 0.25 points (66.36 ± 0.18 vs 66.11 ± 0.02), roughly one standard deviation of the first-token runs. With two seeds per arm this is **no meaningful difference**, and the error profiles match as well (89.5% vs 89.4% of errors are adjacent, Table 4). Our reading: when all 124 M parameters are fine-tuned on 200k examples, the encoder learns to put the review-level information wherever the head reads it, so the `<s>` vector becomes as good a summary as the average. The one consistent difference is stability: mean pooling varied 2–12 times less between seeds on every metric, plausibly because averaging over all tokens smooths out seed-dependent idiosyncrasies of a single position. The brief's choice of mean pooling is therefore not more accurate here, but it is the more reproducible of the two.

7 Ordinal structure: adjacent vs distant errors

Model	% of test reviews			Confusions between two ratings (both directions, per ru					
	off by 1	off by ≥2	adjacent / all errors	1★↔2★	2★↔3★	3★↔4★	4★↔5★	1★↔3★	1★↔5★
BiGRU + attention	30.89	5.81	84.2%	439.0	404.0	313.5	388.0	155.5	11.5
RoBERTa, mean	30.12	3.57	89.4%	407.0	429.0	291.5	378.5	99.5	5.5
RoBERTa, first token	29.93	3.53	89.5%	404.0	416.0	290.0	386.5	97.0	6.5

Table 4: How far off the errors are (mean of two seeds; 5,000 test reviews per run).

Both models treat the ratings as a scale even though plain cross-entropy knows nothing about order: most mass sits on the diagonal and its neighbours (Fig. 3), and 84% (BiGRU) and 89% (RoBERTa) of all errors are only one

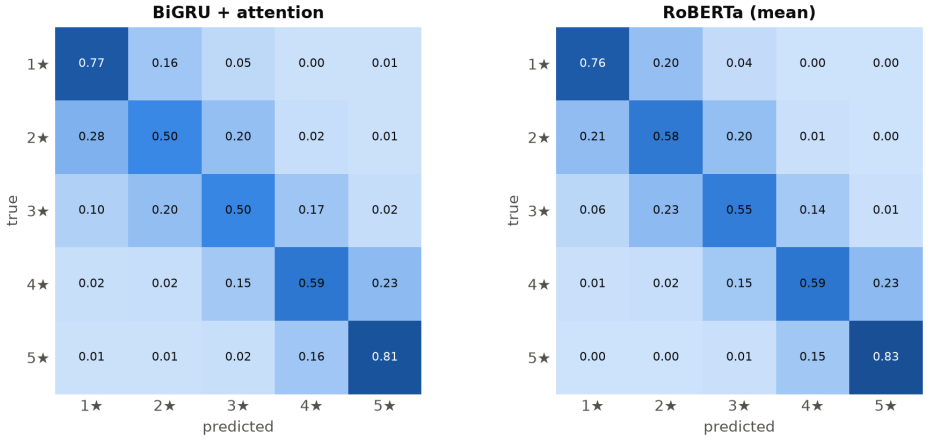


Figure 3: Row-normalised test confusion matrices, counts pooled over both seeds: cell (i, j) is the share of true i-star reviews predicted as j stars.

star off. The difference between the models lies almost entirely in the *distant* errors. RoBERTa's 3.0-point gain in exact predictions comes with a 2.2-point drop in errors of two or more stars (5.81% → 3.57%, a 39% relative reduction) but only a 0.8-point drop in one-star errors; 1★↔3★ swaps fall by a third and 1★↔5★ swaps halve (11.5 → 5.5 per run). Adjacent boundaries, in contrast, barely move: both models send 23% of true 4★ reviews to 5★, 4★↔5★ confusions drop only from 388 to 379, and 2★↔3★ confusions even rise slightly. RoBERTa reads the polarity and rough intensity of a review much better, but where to draw the line between neighbouring ratings is partly a matter of each reviewer's personal scale, which no text model can fully recover. An ordinal loss (e.g. cumulative-link or expected-cost training) would target exactly these boundaries.

8 Qualitative comparison

On the test set, 242 reviews are classified correctly by both RoBERTa seeds and wrongly by both BiGRU seeds; 158 the other way round. Requiring both seeds to agree keeps out examples that are one seed's luck. To see which kinds of review drive the difference we flag four phenomena with simple word patterns (Table 5); the flags are rough and sarcasm cannot be caught by a pattern, so the examples below were read and labelled by hand.

Group (pattern)	Reviews	BiGRU acc. (%)	RoBERTa acc. (%)	Gain
negation (*not, no, never, n't, ...*)	2,858	60.1	63.5	+3.4
mixed / contrast (*but, however, although, ...*)	1,905	56.6	60.3	+3.7
long (>120 spaCy tokens, about the 95th pct.)	216	57.2	60.0	+2.8
rare words / typos (≥2 words outside the BiGRU vocabulary)	158	59.8	66.1	+6.3
none of the above	1,539	70.1	72.8	+2.7

Table 5: Test accuracy (mean of two seeds) by flagged group; a review can be in several groups.

Every flagged group is harder than unflagged reviews for both models, and RoBERTa gains most where its byte-level BPE helps (rare words and typos, +6.3, though on only 158 reviews) and on contrastive reviews (+3.7).

Where RoBERTa wins. The common thread is *composition*: the rating depends on how words combine, not on which words occur. In A and B the negative words are negated or resolved ("don't have cravings", "not to bad", "isn't fully covered ... but most is"); the BiGRU sees "don't" and "horrible" and predicts 1★–3★. C hinges on a product name that is `<unk>` for the BiGRU ("Senseo") and on a comparison ("better than"); E describes a problem at length ("cooked kind of smell") before the product solves it. D is the sarcasm-like case: the ironic "well, let's just say ... deus ex machina" wraps praise for the characters, and RoBERTa settles on the middle rating while the BiGRU follows the positive words. In F the reviewer even states "4.5/5" inside a pros-and-cons list; the two BiGRU seeds disagree wildly (2★ and 5★), RoBERTa reads it correctly.

Where the BiGRU wins. All three are 1★ reviews that RoBERTa rates higher (by one star in five of the six predictions), and each contains a concession: "probably very good", "probably deserves 5", "a decent camera", or in G a complaint about policy rather than the product. RoBERTa weighs these hedges, while the reviewer's verdict is harsh; the BiGRU keys on strongly negative words ("cheating", "useless", "won't eat") and lands on 1★. Its wins are thus mostly adjacent-rating decisions where a cruder reading happens to match how the reviewer mapped sentiment to stars.

	Review (excerpt)	True	BiGRU	RoBERTa	Phenomenon
RoBERTa right, BiGRU wrong (both seeds)					
A	"I don't have cravings after taking this at night! I'm horrible with eating late at night, [...] But when I take this, the cravings are gone!!"	5	1, 2	5, 5	negation, contrast
B	"Fit my 2018 Accord pretty good the back part of the seat isnt fully covered completely but most is, Came a bit wrinkled but not to bad, material is actually good quality"	4	3, 3	4, 4	typos, negated negative
C	"better than the original decaf Senseo pods. [...] almost indistinguishable from the regular caffeinated coffee. Recommend."	5	4, 4	5, 5	rare word, comparison
D	"[...] Whatever that was, it wasn't really present in the sequel. [...] well, let's just say it's all a bit rushed and bordering on repeated examples of deus ex machina [...]"	3	4, 4	3, 3	irony, mixed, long
E	"Works 100% on curling-ironed hair. [...] it tends to leave my hair with a cooked kind of smell. This spray works! [...] it doesn't dry it out" (128 words)	5	3, 2	5, 5	long, problem–solution
F	"The Good: -came with a free gym towel [...] The Bad -the only issue [...] Overall: 4.5/5"	4	2, 5	4, 4	pros/cons list, explicit score
BiGRU right, RoBERTa wrong (both seeds)					
G	"Not fully described. Amazon did't disclose the return policy of restock fee which is called cheating."	1	1, 1	2, 2	short, negation
H	"Probably very good, but my boy won't eat them. I hate to give this product one star when it probably deserves 5. [...]"	1	1, 1	3, 2	concession, hedging
I	"[...] if you're comfortable with being stuck with a subscription every year at $100 per camera then it is a decent camera but it's useless without the subscription."	1	1, 1	2, 2	concession, long

Table 6: Test examples chosen by `pick_examples` (most confident first, covering as many flags as possible); predictions are listed for seeds 42, 43.

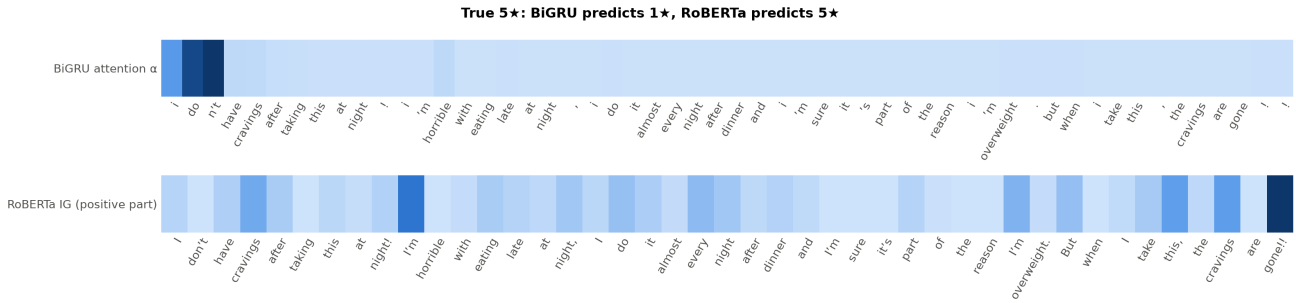


Figure 4: Example A. Top: BiGRU attention weights α_t (sum to 1). Bottom: RoBERTa token importance by integrated gradients for the predicted 5★ logit, sub-word pieces summed into words; the positive part is shown, normalised to sum to 1. Each row has its own colour scale.

What the models look at. For RoBERTa we use integrated gradients (Sundararajan et al., 2017) on the input embeddings: 256 steps from a baseline in which every token except `<s>` and `</s>` is replaced by `<pad>`. The attributions add up to the change in the logit (here 4.24, with a completeness error of 0.04). In Fig. 4, the strongest evidence *for* 5★ is "gone!!" at the very end (+1.09) and the first-person "I'm" (+0.65); "don't" (−0.62) and "horrible" (−0.41) are the strongest evidence *against*. RoBERTa therefore registers the negative words as negative, but the resolution at the end outweighs them. The BiGRU's attention, by contrast, piles onto "do" and "n't" in the first sentence and it predicts 1★: the negation cue is found, but the pooled vector carries "negative" rather than "negated negative". A second example (`test_token_importance_1.png`, example C) shows both models focusing on "Recommend" (IG +1.90), yet only RoBERTa reaches 5★; the BiGRU also has to read "Senseo" as `<unk>`. Two caveats apply: attention weights show what was pooled, not what changed the decision, and integrated gradients depend on the chosen baseline.

9 Conclusions and limitations

- Fine-tuned RoBERTa is clearly better than a well-tuned BiGRU + attention baseline: +3.3 macro-F1 and −0.059 MAE on test, ahead in every class, with the largest gains on the ambiguous middle ratings.
- The gain is mostly about *severity*: RoBERTa removes 39% of the errors of two or more stars and half of the 1↔5 swaps, but adjacent-rating confusions (especially 4★ vs 5★) stay at the baseline's level.
- Qualitatively, RoBERTa handles negated negatives, contrast, rare words and typos better; its own errors are concessive 1★ reviews that it rates too high, usually by one star.
- Mean and first-token pooling perform the same within seed noise; mean pooling is the more stable choice.

Limitations. (i) Two seeds give only a rough estimate of variance, so small differences such as the pooling ablation cannot be resolved. (ii) Every RoBERTa run peaked at its final checkpoint; a third epoch might help, but retraining after the test predictions existed would have meant evaluating the test set twice, so we report the runs as they are. (iii) Confusion matrices pool both seeds, and the phenomenon flags are word patterns, not annotations. (iv) The BiGRU overfits quickly after its embeddings are unfrozen; a smaller learning rate for the embedding might give it a fairer chance. (v) Truncation affects under 1% of reviews and is not a factor here.

Reproducibility. Code: `Question-2/` (notebooks `01`–`04` and package `q2/`); each run writes its configuration, seed, library versions and git commit to `run_info.json`. Training ran on Kaggle (one T4 for the BiGRU, two T4s for RoBERTa) with PyTorch 2.11 and `transformers` 5.21.0.